

# CONSTANT OF PROPORTIONALITY – PART 2



## LESSON INTRODUCTION

**MA.7.AR.4.2** Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

## LEARNING TARGETS

- I can determine the constant of proportionality from a written description.
- I can relate the constant of proportionality to an equation representing a proportional relationship.

## BENCHMARK COHERENCE

### BUILDING ON

**MA.6.AR.3.2** Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

### WORKING TOWARD

**MA.8.AR.3.2** Given a table, graph, or written description of a linear relationship, determine the slope.

**MA.8.AR.3.4** Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.

## ESSENTIAL QUESTIONS

- What is the constant of proportionality?
- How can you use a graph, table, or written description to determine a constant of proportionality?
- How do you interpret the constant of proportionality in context?

## LESSON NARRATIVE

This lesson builds on prior knowledge of using graphs and tables to recognize proportional relationships. Students will continue developing their understanding of constant of proportionality. Students will then use graphs, tables, and written descriptions to determine and interpret the constant of proportionality in several real-world contexts. Students will also begin exploring how to express these relationships using equations.

## COMPONENT SUMMARY

### 11.3.1 WARM-UP (~5 MIN.)

Describe a situation that could be described by a given graph

### 11.3.3 GUIDED PRACTICE (5 MIN.)

Identify the constant of proportionality in real-world context

### 11.3.5 WRAP-UP (~5 MIN.)

Reflect on the understanding proportional relationships

### 11.3.2 GUIDED INSTRUCTION (5-20 MIN.)

Find and interpret the constant of proportionality from a graph or table of values

### 11.3.4. COLLABORATION (10 MIN.)

Represent proportional relationships with equations

## RESOURCES

### GLOSSARY

Key Terms:

- Constant of proportionality

Additional Terms:

- Ratio

### MATERIALS

- Calculator
- (Optional) Colored pencils or highlighters

### ADDITIONAL PREPARATION

- N/A

## TEACHER TIPS

### COMMON MISCONCEPTIONS

- Students may struggle to understand the two constants of proportionality in context and may mix up which constant of proportionality represents the corresponding direction of the proportional relationship.

### THINGS TO CONSIDER

- Consider encouraging students to verbalize the meaning of each constant of proportionality in context as they complete each section. This can aid students in understanding written descriptions when given without tables of values or tables.

### LITERACY AND LANGUAGE ROUTINES

#### Think-Pair-Share

11.3.2 Guided Instruction provides an opportunity to engage students in a Think-Pair-Share routine as they consider whether the mathematical representation of the graph makes sense within the real-world context of the problem.

### MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

#### MA.K12.MTR.5.1 Patterns and Structure

In 11.3.4 Collaboration, students begin considering how to represent a proportional relationship using an equation. They are positioned to use the patterns observed in the table of values to relate one variable to another as they determine which constant of proportionality to use in each equation depending on which variable is being used as the independent variable.

## DIFFERENTIATE INSTRUCTION

This lesson provides a variety of visual and auditory entry points. Graphs and tables are used to represent proportional relationships as students consider what the constant of proportionality means in context. Students collaborate with partners to complete the work in each section, encouraging auditory and verbal processing through discussion.

Encourage students to use colors to label the x- and y-values in each context to support them in their work determining the constant of proportionality.

## 11.3.1 WARM-UP: BELL WORK

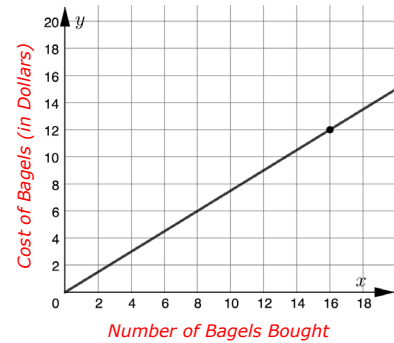
### DIFFERENTIATE INSTRUCTION

As students work, circulate and observe how they are completing each question. Consider using informal observations from this section to inform small groups or partner work during the 11.3.4 Collaboration.



### 11.3.1 Warm-Up: Bell Work

The graph represents a proportional relationship.

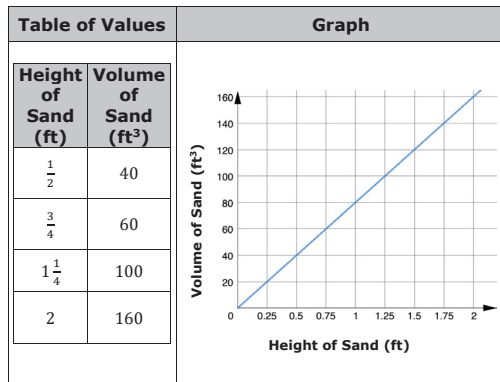


- Describe a situation that could be represented by this graph.  
*Answers will vary.*  
*The graph could describe the price per number of bagels bought, with 1 bagel costing \$0.75.*
- Label the axes with the quantities in your situation.  
*Labels will vary, depending on context given in #1.*
- There is a point marked at (16,12) on the graph. What does this point represent in your situation?  
*Answers will vary, depending on context given in #1.*  
*The point (16,12) represents that 16 bagels will cost \$12.*



### 11.3.2 Guided Instruction: Relating the Constant of Proportionality to Written Descriptions

- The volume of sand in a rectangular sandbox,  $y$ , changes proportionally with relationship to the height of the sand in the sandbox,  $x$ . The table of values and graph represent this relationship for a rectangular sandbox measuring 10 feet (ft) long and 8 ft wide with the capacity to fill to a height of 2 ft.



- Determine the constant of proportionality,  $\frac{y}{x}$ , using the table of values. Show your work.  

$$\frac{40}{0.5} = \frac{60}{0.75} = \frac{100}{1.25} = \frac{160}{2} = 80$$

The constant of proportionality is 80.

## 11.3.2 GUIDED INSTRUCTION: RELATING THE CONSTANT OF PROPORTIONALITY TO WRITTEN DESCRIPTIONS

### TEACHER MOVES

There are three ways to determine the constant of proportionality from a graph of a proportional relationship:

- Identify the point along the graph where  $x=1$ . The  $y$ -value will be the constant of proportionality. This point is not always easily identifiable, but when it is, this strategy can be used. If students explain this strategy here, consider asking what they might do if the value of  $y$  when  $x=1$  is not easily identifiable from the graph.
- Choose any point along the graph (other than the origin) and find the ratio  $\frac{y}{x}$  of the coordinates. Because this ratio is equivalent at every point on the graph of a proportional relationship, any point that can be identified will work.
- Find the change in  $y$  over the change in  $x$  between any two points on the graph. Because students have not been introduced to slope yet (and that is not the intention of these benchmarks), it is unlikely that students would make this selection. However, teachers could show this relationship as a preview of slope and slope triangles if it seems appropriate. The first two strategies suffice for Grade 7 proportional relationships.

## 11.3.2 GUIDED INSTRUCTION: RELATING THE CONSTANT OF PROPORTIONALITY TO WRITTEN DESCRIPTIONS (CONTINUED)

### DIFFERENTIATE INSTRUCTION

To provide multiple means of representation, consider projecting the graph and table from question 1 for the class to see. Displaying the information in a flexible format (digitally projected) allows for ways to alter the visuals to provide entry points for all students (i.e.: enlarging the font to make the scales visually accessible).

### ENRICHMENT

This is the first-time students will be working with a constant of proportionality that has a negative value, which may be confusing to some. Pose questions to students to draw attention to the graph and utilize the visual to determine the constant of proportionality.

- What do you notice about the direction of the line representing the proportional relationship? Does this graph look like the other graphs we have worked with so far in this unit?
- What is different about it? What do you think that means for the constant of proportionality?
- What does a negative constant of proportionality mean in context?

- b. Explain how to determine the constant of proportionality using the graph.

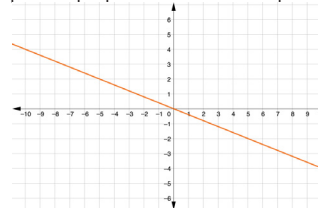
*Answers will vary.*

*Find the value of  $y$  when  $x = 1$  to find the constant of proportionality if it is clear from the graph. Use any point on the graph and find the value of  $\frac{y}{x}$ .*

- c. Describe the meaning of the constant of proportionality in this context.

*The volume of sand in the sandbox increases by  $80 \text{ ft}^3$  as the height of the sand in the sandbox increases by 1 ft.*

2. The graph of a proportional relationship is shown.



- a. Complete the table of values to represent the same relationship.

| $x$ | -5 | -3  | 0 | 1    | 5  | 6.2   |
|-----|----|-----|---|------|----|-------|
| $y$ | 2  | 1.2 | 0 | -0.4 | -2 | -2.48 |

- b. Verify the values in your table with your partner and resolve any differences, if necessary.
- c. What is the constant of proportionality,  $\frac{y}{x}$ , in this relationship?

*The constant of proportionality is  $-\frac{2}{5}$  or  $-0.4$ .*

- d. Verify the ratio of  $\frac{y}{x}$  is constant for each coordinate pair represented in your table using a calculator. Make edits to your work as needed.

$$\frac{2}{-5} = -0.4 \quad \frac{1.2}{-3} = -0.4 \quad \frac{-0.4}{1} = -0.4$$

$$\frac{-2}{5} = -0.4 \quad \frac{-2.48}{6.2} = -0.4$$

- e. Discuss with your partner how the constant of proportionality can be used to determine the  $y$ -values corresponding to specific  $x$ -values in the table in cases where it is difficult to determine from looking at the graph.
- f. This proportional relationship can be used to represent the relationship between the original price,  $x$ , and the change in price due to a store-wide sale,  $y$ . Complete the statement to interpret the constant of proportionality found in Part C in this context.

The constant of proportionality is -0.4, so the

change in price due to the sale is a 40%

- ☐ reduction  
☐ mark up

from the original price. This means

the store is having a 40% off sale.

- g. Explain which part(s) of the graph representing this proportional relationship would not make sense mathematically for this real-world context.

*The  $x$ -axis would represent the original cost of items, and the  $y$ -axis would represent the value of the discount. Therefore, the values graphed in Quadrant II would not make sense, because the  $x$ -value (the original price) will always be a positive value.*

## 11.3.2 GUIDED INSTRUCTION: RELATING THE CONSTANT OF PROPORTIONALITY TO WRITTEN DESCRIPTIONS (CONTINUED)

### TEACHER MOVES

Question 2e is meant to lead toward the equation  $y=kx$ , but students do not need to write an equation. Rather, it's an opportunity for them to consider that the  $y$ -value can be determined by multiplying  $y/x$  by  $x$ .

### TEACHER MOVES

#### Think-Pair-Share

Engage students in a Think-Pair-Share routine as they complete part g. Encourage students to critique and build upon their partner's thinking as they verbalize their responses. Circulate as students share with their partners and choose one to two pairs to share out. Highlight students who are connecting the axes on the graphs to their real-world representation.

### 11.3.3 GUIDED PRACTICE: DETERMINING THE CONSTANT OF PROPORTIONALITY FROM A WRITTEN DESCRIPTION

#### TEACHER MOVES

Some students may still mix up the x- and y-values when determining the constant of proportionality. Encourage students to label the values with x or y or use other visual cues to keep track, such as color-coding or circling and underlining.



#### 11.3.3 Guided Practice: Determining the Constant of Proportionality from a Written Description

1. Kyla is training for a half-marathon. She sets her treadmill to a constant speed to build up her endurance on longer runs. The relationship between the time she runs,  $x$ , and the distance she runs,  $y$ , is proportional. She runs 10.5 miles in 1.75 hours at this pace.
  - a. Determine the constant of proportionality,  $\frac{y}{x}$ .  
*The constant of proportionality is 6.*
  - b. Complete the statement to interpret the constant of proportionality.

At this rate, Kyla runs 6 
☐ miles  
☐ hours
  for every 
☐ 1 mile  
☐ 1 hour
  she trains on the treadmill.

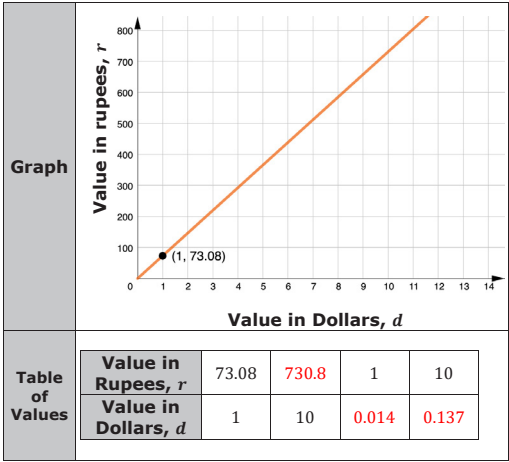


In a real-world context, the **constant of proportionality** is the amount one quantity increases or decreases each time the other quantity increases by 1 unit.



11.3.4 Collaboration: Equations Representing Proportional Relationships

On January 14, 2021, 1 U.S. dollar was worth 73.08 Indian rupees. This is a written description of a proportional relationship. Below are two other representations of the same relationship.



- Work with your partner to complete the following.
- Complete the table of values by filling in the missing values. Round values to the nearest thousandth, where necessary.

11.3.4 COLLABORATION: EQUATIONS REPRESENTING PROPORTIONAL RELATIONSHIPS



## 11.3.4 COLLABORATION: EQUATIONS REPRESENTING PROPORTIONAL RELATIONSHIPS (CONTINUED)

### TEACHER MOVES

#### MA.K12.MTR.5.1: Patterns and Structure

Encourage students to use the patterns in the table to determine which unit rate, or constant of proportionality, can be used to relate the variables.

- What value multiplied by  $r$  results in  $d$  for each coordinate pair?
- What value multiplied by  $d$  results in  $r$  for each coordinate pair?

### ENRICHMENT

As students explore how to create equations to represent proportional relationships, provide guidance through questioning.

- What is the constant of proportionality when describing the number of rupees per 1 U.S. dollar?
  - Are there more rupees per dollar or vice versa? How can you show this using an equation?
- What is the constant of proportionality when describing the number of U.S. dollars per 1 rupee?
  - Are there more dollars per rupee or vice versa? How can you show this using an equation?

### TEACHER MOVES

Equations will come back up in future lessons with more direct instruction. Question 5 is an opportunity for students to discuss and explore how the relationships they are working with can be represented by equations and how the constant of proportionality relates in each equation.

Proportional relationships can also be represented by equations.

2. Use the quantities determined in the table of values to complete each equation that represents this relationship.

$$d = 0.014 \cdot r \quad \text{or} \quad r = 73.08 \cdot d$$

3. What is the constant of proportionality,  $\frac{d}{r}$ , in this context?  
*The constant of proportionality,  $\frac{d}{r}$ , is 0.014. Every 0.014 US dollar is equivalent to 1 rupee.*
4. What is the constant of proportionality,  $\frac{r}{d}$ , in this context?  
*The constant of proportionality,  $\frac{r}{d}$ , is 73.08. Every 1 US dollar is equivalent to 73.08 rupees.*
5. Discuss with your partner how the constants of proportionality relate to the equations of the proportional relationship.



### 11.3.5 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding of the benchmark below.

Determine the constant of proportionality within a mathematical or real-world context given one of these forms: a table, graph, or written description of a proportional relationship.



I'm so confident, I could explain how to determine the constant of proportionality for proportional relationships to someone else!

I can get to the right answer, but I don't understand well enough to explain how to determine the constant of proportionality yet.

I understand how to find the constant of proportionality for some representations of proportional relationships, but I don't understand how to for all forms yet.

I have been trying hard, but I am not yet able to determine the constant of proportionality for proportional relationships.

I do not understand how to determine the constant of proportionality for proportional relationships yet. There are things I could do to be a better learner next lesson.

*Answers will vary.*

## 11.3.5 WRAP-UP: REFLECTION

### DIFFERENTIATE INSTRUCTION

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.